

Toby Rodel - CV & Publication list

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EDUCATION

October 2023 - March 2027	PhD (Astrophysics) at Queen's University Belfast	(In progress)
October 2022 - October 2023	MSc by Research (Physics) at University of Warwick	(Completed)
October 2019 - June 2022	BSc (Physics) at University of Warwick	(2:1)

RESEARCH EXPERIENCE

PhD (Queen's University Belfast with Prof Christopher Watson) 2023 - March 2027 (est)

- Characterisation of long-period transiting planets and eclipsing binary systems including NGTS-EB-7 (Rodel et al. [2025](#)) and TIC-65910228 / NGTS-38 (Rodel et al. [2026](#)).
- Used PAWS/iSpec and isochrones to characterise host stars and Allesfitter to fit orbital solutions to transit and radial velocity data.
- Submitted succesful ESO proposal for multiband transit observations with ULTRACAM for validation.
- Continuation of MSc work predicting survey yields, including upgrades to custom pipeline.

MSc by research (University of Warwick with Dr Daniel Bayliss) 2022 - 2023

- Developed the Transit Investigation and Recoverability Application (TlARA) and used it to predict exoplanet discovery yields from *TESS*-SPOC data (Rodel et al. [2024](#)).
- Participated in eyeballing and vetting of *TESS* monotransit candidates.

BSc project (University of Warwick with Dr Matteo Brogi) 2021-2022

- Developed code to detect and fit secondary eclipses of known hot Jupiters in *TESS* data.
- Attempted albedo and heat redistribution factor determination.

SKILLS & EXPERTISE

Research expertise: exoplanets, transits, radial velocities, wide field photometric surveys, long-period, low mass eclipsing binaries, stellar characterisation, survey yield prediction, occurrence rates.

Instruments: *TESS*, NGTS, ULTRACAM, CORALIE, HARPS, FEROS, LCOGT

Software: TlARA (custom pipeline), [Allesfitter](#), [PAWS](#), [iSpec](#), [isochrones](#), [Specmatch-emp](#), [AstroAriadne](#)

Programming languages: Python, SQL, HTML, CSS, $\text{\LaTeX}$, markdown

Other: public speaking and engagement, poster design, mentoring, acronym invention

SERVICE & LEADERSHIP

ESO DPR reviewer P114 & P115

NGTS communications chair 2023 - 2025

- First chair of NGTS communications group to manage consortium promotion and outreach.
- Managed consortium social media and maintained webpage.
- Procured customised pins and stickers for consortium members.

QUB Equitea committee member 2023 - present

- Planned seminars and discussion sessions on Equity, Diversity and Inclusion topics and their relations to Astronomy

QUB ARC seminar committee member

2024-2025

- Arranged for external speakers to give seminars and hosted them.

TEACHING AND MENTORING

Co-supervisor of summer intern

June-September 2026

- Helped design cross-disciplinary project extracting SNe lightcurves from NGTS data.
- Sat on panel to shortlist applicants.
- Supervised student in developing pipeline to extract photometry from NGTS images.

Postgraduate teaching assistant - Queen's University Belfast

2023-2024

- Problem class tutor for modules MTH1021 and PHY1002.

Mentoring of research students

2023 - Present

- Assisted and mentored QUB MSci students with software, proofreading of reports and presentation practice.
- Assisted and mentored external students through the NGTS early careers group.

OUTREACH

Planetarium programmes

2022 - Present

- Setup and take-down of inflatable dome as well as presentation of shows using Stellarium and Universe Sandbox at various festivals and school visits.
- Volunteered at Warwick, cofounded new programme in Belfast.

Festivals and events

Various

- **QUB Astronomy day: Volunteer 2024 & 2025**, on organising committee for 2026.
- **Astronomy on Tap Belfast** Speaker and Organiser / MC.
- **Big Bang Science Festival Birmingham**: Volunteer at Warwick booth.

Public talks and Lectures

Various

REFEREES

Prof Christopher Watson	PhD supervisor - Queen's University Belfast	c.a.watson@qub.ac.uk
Dr Daniel Bayliss	MSc supervisor - University of Warwick	d.bayliss@warwick.ac.uk
Dr Sarah Casewell	Collaborator - University of Leicester	slc25@leicester.ac.uk
Dr Ernst de Mooij	PhD 2nd supervisor - Queen's University Belfast	e.demooij@qub.ac.uk
Dr Matteo Brogi	BSc supervisor - University of Turin	matteo.brogi@unito.it

FIRST AUTHORED PUBLICATIONS

- Rodel, Toby et al. (Feb. 2025). “NGTS-EB-7, an eccentric, long-period, low-mass eclipsing binary”. In: MNRAS 537.1, pp. 35–55. DOI: [10.1093/mnras/stae2799](https://doi.org/10.1093/mnras/stae2799).
- Rodel, Toby et al. (July 2026). “TIC-65910228 b/NGTS-38 b, a 180 day transiting warm super-Jupiter”. In: MNRAS 549.4, stag1061. DOI: [10.1093/mnras/stag1061](https://doi.org/10.1093/mnras/stag1061).
- Rodel, Toby et al. (Mar. 2024). “TIARA TESS 1: estimating exoplanet yields from Years 1 and 3 SPOC light curves”. In: MNRAS 529.1, pp. 715–731. DOI: [10.1093/mnras/stae474](https://doi.org/10.1093/mnras/stae474).

CO-AUTHORED PUBLICATIONS

- Mallaghan, Niamh et al. (Sept. 2026). “NGTS clusters survey - VII. An Enigmatic Short-Period Circum-secondary Disk Candidate in Orion”. In: MNRAS. DOI: [10.1093/mnras/stag1802](https://doi.org/10.1093/mnras/stag1802).
- Hobbs, Katlyn L. et al. (Aug. 2026). “The Fe I 4377 Å Line as a Solar Faculae Indicator: Insights from Spectral Ratio Analysis”. In: MNRAS. DOI: [10.1093/mnras/stag1462](https://doi.org/10.1093/mnras/stag1462).
- Apergis, Ioannis et al. (Aug. 2026a). “NGTS-39 b: a 58 d transiting warm Jupiter in an eccentric orbit”. In: MNRAS 550.2, stag1271. DOI: [10.1093/mnras/stag1271](https://doi.org/10.1093/mnras/stag1271).
- Kendall, Alicia et al. (Apr. 2026). “A 43 d transiting Neptune and two 25 d Saturns from TESS, NGTS, and ASTEP”. In: MNRAS 547.2, staf2189. DOI: [10.1093/mnras/staf2189](https://doi.org/10.1093/mnras/staf2189).
- Apergis, Ioannis et al. (Jan. 2026b). “High-precision photometry with a scientific CMOS camera: II on-sky testing of the Marana camera at the NGTS facility”. In: *RAS Techniques and Instruments* 5, rzag022. DOI: [10.1093/rasti/rzag022](https://doi.org/10.1093/rasti/rzag022).
- O’Brien, Sean M. et al. (Dec. 2025). “NGTS-EB-8: A Double-lined Eclipsing M+M Binary Discovered by Citizen Scientists”. In: AJ 170.6, p. 316. DOI: [10.3847/1538-3881/ae0e0d](https://doi.org/10.3847/1538-3881/ae0e0d).
- Ulmer-Moll, S. et al. (Nov. 2025). “Detection and characterisation of a 106-day transiting Jupiter: TOI-2449 b/NGTS-36 b”. In: A&A 703, A258. DOI: [10.1051/0004-6361/202555168](https://doi.org/10.1051/0004-6361/202555168).
- Eschen, Yoshi Nike Emilia et al. (Dec. 2024). “Viewing the PLATO LOPS2 field through the lenses of TESS”. In: MNRAS 535.2, pp. 1778–1795. DOI: [10.1093/mnras/stae2427](https://doi.org/10.1093/mnras/stae2427).
- Doyle, Lauren et al. (Apr. 2024). “The TESS-SPOC FFI target sample explored with Gaia”. In: MNRAS 529.2, pp. 1802–1813. DOI: [10.1093/mnras/stae616](https://doi.org/10.1093/mnras/stae616).
- Hawthorn, Faith et al. (Feb. 2024). “TESS duotransit candidates from the Southern Ecliptic Hemisphere”. In: MNRAS 528.2, pp. 1841–1862. DOI: [10.1093/mnras/stad3783](https://doi.org/10.1093/mnras/stad3783).

OBSERVING PROGRAMMES

As PI

“Probing puffy exoplanet atmospheres with ULTRACAM” ESO, 118.25EL, ULTRACAM on NTT (19h)

As Co-I

“Characterising transiting warm and temperate Jupiters to shed light on their formation” HARPS-N ITP 2026B & 2027A, HARPS-N (15N), RISE (70h), HiperCam (11h)

CONFERENCE PRESENTATIONS

Talks

- “Finding hidden long-period planets with *TESS*”, *UK Exoplanet Meeting*, University of Bristol, April 2026
- “How good is *TESS* at detecting long-period exoplanets”, *From Transits to Trends: the Next Decade of Long-Period Exoplanets*, University of New Mexico, August 2025
- “Long period planets from *TESS* and NGTS”, *Irish National Astronomy Meeting*, University of Galway, August 2024
- “Putting a TIARA on SPOC”, *TESS Science Conference 3*, Massachusetts Institute of Technology, August 2024

Posters

- “Eclipsing Binaries from the Next Generation Transit Survey”, *National Astronomy Meeting*, University of Birmingham, July 2026
- “Finding hidden long-period planets with *TESS*”, *National Astronomy Meeting*, University of Birmingham, July 2026
- “Finding hidden long-period planets with *TESS*”, *Exoplanets 6*, Instituto de Astrofísica e Ciências do Espaço, July 2026
- “NGTS-EB-7: One of the longest period EBLM systems ever found”, *Binary stars in the space era*, University of Keele, July 2025
- “NGTS-EB-7: One of the longest period EBLM systems ever found”, *UK Exoplanet Meeting*, University of Leeds, April 2025
- “Putting a TIARA on SPOC”, *TESS Science Conference 3*, Massachusetts Institute of Technology, August 2024
- “Putting a TIARA on SPOC: Predicting long-period planet yields from *TESS*”, *UK Exoplanet Meeting*, University of Birmingham, April 2024
- “Predicting the yield of long-period Planets in *TESS*”, *UK Exoplanet Meeting*, University College London, August 2023